

FREE ELECTRIC VEHICLE ENGINEERING WORKSHOPS



The Lab **Boise**

For Students, Grades 1 through 12

Using a real student-built electric vehicle
as a hands-on example

LEARN MORE:
thelabboise.org

About 1 hour | No prior engineering experience required

FROM GAS-POWERED CHASSIS



TO STUDENT-BUILT ELECTRIC VEHICLE



1 WORKSHOP 1 UNDERSTAND IT

How an EV Works

- EV components and how they connect
- Basic electrical and mechanical concepts
- Energy, torque, and drivetrain basics

2 WORKSHOP 2 PLAN IT

How an Engineering
Project Takes Shape

- How to start an engineering project
- Planning, decision-making, and troubleshooting
- Lessons learned from the EV build

3 WORKSHOP 3 DESIGN IT

Introduction to CAD

- CAD basics for beginners
- How engineers use CAD to plan parts
- Introductory design concepts



ABOUT THE PRESENTER

Vibodh is a Timberline high school senior who loves technical engineering challenges and has led the electric vehicle project featured in the workshops. He has also won awards at the 2026 RTX National Invention Convention for his technical innovation.

SCAN TO REGISTER

